Metadata for Trofarth (TRO), Conwy, UK

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| Description of data set | Magnetic time-series from the BGS SWIGS differential magnetometer method (DMM) systems. Funded by NERC, grant number: NE/P017231/1 "Space Weather Impact on Ground-based Systems (SWIGS)". These data consist of measurements of the Earth's natural magnetic field at the remote site (TROR) and the natural magnetic field plus the field created by GIC at the underline site (TROU). The database will include .xyz files with the DMM data and one document with metadata. See Hübert et al. (2020) for further details. |
| Location (in lat/long): underline station | 53.22362 N, -3.7144 E |
| Location (in lat/long): remote station | 53.22537 N, -3.71513 E |
| Distance between DMM sites | 200 m |
| Description of site (i.e. open moorland) | Field in rolling hills, close to drystone wall |
| Equipment used | Sensys 3DFM fluxgate magnetometer and Earth Data digitisers (EDR 209, 24bit) |
| Installation date | 11/08/2021 |
| Removal date | 30/03/2022 |
| Number of days collected | TROR: 77<br>TROU: 172 |
| Data sampling rate | 1 Hz |
| Notes of known data gaps | TROR: 29/10/2021-30/03/2022<br>TROU: 01/01/2022-17/01/2022, 28/02/2022-30/03/2022 |
| Qualitative estimate of site quality | TROR: very good<br>TROU: very good, some HF noise on Bz |
| Orientation angle of HV wires relative to true north | +22° (i.e. lines run approximately N-S) |
| Alignment of magnetometers | Magnetic north as of 11/08/2021 |
| Voltage level of the lines | 400 kV (Pentyr – Bodelwyddan) |
| Height above the ground of each wire (m) | Lower: 23.09 m, Middle: 31.87 m, Upper: 41.91 m |
| Underline magnetometer | Central (9m to either circuit) |
| File Type | ASCII text (.xyz) |
| File format | YYYY-MM-DD HH:MM:SS X Y Z<br>Five columns with 86400 rows (plus one header row). Magnetic values in units of nT (10^-9 T).<br>Empty values denoted by: 999999.00 |
| Citation | Hübert, J., Beggan, C. D., Richardson, G. S., Martyn, T., & Thomson, A. W. P. (2020). Differential magnetometer measurements of geomagnetically induced currents in a complex high voltage network. Space Weather, 18, e2019SW002421. doi: 10.1029/2019SW002421 |